**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 090 · Combining Boolean Logic

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What are the NOT outputs for input 0 then input 1?

Q2. What does a structure diagram show?
A. The order instructions run
B. Parts and their hierarchy
C. Every variable's value
D. The network's IP addresses

Q3. “Computers use binary because humans cannot write numbers above 1.” Correct this. Explain why.

Q4. A program reads exactly ten scores. Another asks for a password until a condition is met. Correct both claims. Explain why. (a) Count-controlled loops are only for unknown repeat counts. (b) Condition-controlled loops always run ten times.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Combined logic — Several linked gates forming a Boolean expression.

Intermediate output — Result from a gate before the final gate.

AND — True only with two high inputs.

OR — True when at least one input is high.

NOT — Reverses a Boolean value.

Truth table — Every input combination and its output.

## New learning

Build gates in sequence: Combined logic sends one gate's output into another; label each intermediate wire clearly.

Work inside parentheses: For Q=(A AND B) OR C, calculate X=A AND B before finding Q=X OR C.

All three-input rows: With A,B,C, there are eight input combinations: 000,001,010,011,100,101,110,111.

Check intermediate values: Record X and Q separately to spot where a wrong gate evaluation changes the final result.

## Worked example

A circuit is Q=(A AND B) OR C. Find X=A AND B and Q for inputs (1,0,1) and (1,1,0). [4 marks]

1. Work from the AND gate, X=A AND B, then OR X with C.

2. For inputs (1,0,1), X=1 AND 0=0 [1].

3. Q=0 OR 1=1 [1].

4. For (1,1,0), X=1 [1] and Q=1 OR 0=1 [1].

# We Do · Practise with guidance

### Guided 1

For Q=(A AND B) OR C, evaluate inputs 0,1,0 and 1,1,1. Show X and Q. [4 marks]

Worked method: For 0,1,0: X=0 [1] and Q=0 [1]. For 1,1,1: X=1 [1] and Q=1 [1].

### Guided 2

A different system uses Q=NOT A OR B. Find Q when inputs are 1,0 and 0,0. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: For 1,0, NOT A=0 [1] so Q=0 [1]. For 0,0, NOT A=1 [1] so Q=1 [1].

**Guided 3**

Explain why Q=(A AND B) OR C is always 1 when C=1. [4 marks]

Worked method: X=A AND B can be 0 or 1 [1]. OR combines X with C [1]. With C=1, at least one OR input is 1 [1], so Q=1 [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the condition so the alert appears when malware is found or when both login checks fail.

```
unknownDevice = true
repeatedFailures = false
malware = true
if ... then
    print("Investigate")
endif
```

write

### Question 1

State the intermediate expression X for Q=(A AND B) OR C. [1 mark] Source: Original OCR J277-style practice question.

### Question 2

Find X and Q for (A,B,C)=(0,0,1) and (1,1,0). [4 marks] Source: Original OCR J277-style practice question.

### Question 3

Complete the Q column for Q=(A AND B) OR C with inputs 000,001,010,011,100,101,110,111, and explain the 110 output. [6 marks] Source: Original OCR J277-style practice question.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

For Q=(A AND B) OR C, Q=1 whenever C=1.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. 1 then 0.

Q2. B — Parts and their hierarchy

Q3. Digital circuits can reliably distinguish two states, represented by 0 and 1.

Q4. (a) Use a count-controlled loop when the number of repeats is known. (b) Use a condition-controlled loop when a condition decides when it stops. Choose the right loop and trace its steps.

## I Do / We Do pages · Guided practice

Q1. For 0,1,0: X=0 [1] and Q=0 [1]. For 1,1,1: X=1 [1] and Q=1 [1].

Q2. For 1,0, NOT A=0 [1] so Q=0 [1]. For 0,0, NOT A=1 [1] so Q=1 [1].

Q3. X=A AND B can be 0 or 1 [1]. OR combines X with C [1]. With C=1, at least one OR input is 1 [1], so Q=1 [1].

## You Do pages · Code First

```
unknownDevice = true
repeatedFailures = false
malware = true
if (unknownDevice AND repeatedFailures) OR malware then
    print("Investigate")
endif
```

Expected result: Investigate

## You Do pages · Independent questions

Q1. X=A AND B [1].

Q2. For 0,0,1 X=0 [1], Q=1 [1]. For 1,1,0 X=1 [1], Q=1 [1].

Q3. Q column: 0,1,0,1,0,1,1,1 [4 marks: one for each correct adjacent pair]. For 110, A AND B=1 [1], so 1 OR C=0 gives Q=1 [1].

## Exit Ticket · Model answer

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**